Sai Prasanth Parnambedu

Robotics & Software Engineer | AI Integration | Automation Systems

Date of Birth: 12.08.1996, Chennai, India

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PROFILE

Robotics & Software Engineer with 3+ years of experience in **ROS/ROS2, C++, Python, and robotic system integration** (Fraunhofer IPA). I hold a Master's in Mechatronics and am currently pursuing Applied AI for Digital Production Management, aiming to bridge artificial intelligence with industrial robotics. My work focuses on trajectory validation, force-controlled manipulation, and machine learning–driven automation for smarter, adaptive robotic systems.

EDUCATION

March 2025 – Ongoing

M.Eng. Applied AI for Digital Production Management (Grade: 2.1)
Technische Hochschule Deggendorf, Germany
Modules: Production Data Management, ML Algorithms, Statistical Methods

October 2019 – August 2023

M.Sc. Mechatronics (Grade: 2.5)
Ravensburg Weingarten University, Germany
Modules: Robotics, Computer Vision, Control Systems

July 2014 – May 2018

B.Tech. Mechanical Engineering (Grade: 1.8)
SRM Institute of Science and Technology, India
Modules: Finite Element Methods, Computer Aided Design and Manufacturing

WORK EXPERIENCE

April 2024 – Present
Freelance / Remote

Independent Robotics & AI Engineer

- **Smart inspection cell**: Trained YOLOv8 and MobileNetV2 vision models on synthetic dataset, integrated into RoboDK simulated robotic cell to compare inference latency and productivity rate for automotive component defect detection.
- **WeDoWind Challenge**: Built ML pipelines for wind turbine health monitoring; using rolling variance with isolation forest, achieving **F1-score > 0.85** in fault detection.
- Published and maintained open-source robotics and ML repositories (Tiago Robot Project, Sentiment Analysis Project), demonstrating applied skills in ROS, MoveIt, and ML (scikit-learn, TensorFlow) with detailed documentation.

December 2021 – August 2023
Fraunhofer IPA, Stuttgart, Germany

Working Student in Robotics

- Validated robot trajectories with rosbag, FastDTW and Python, improving accuracy verification.
- Integrated trajectory tests into GitLab CI/CD, enabling continuous validation and reducing regression errors by **25%**.
- Commissioned **3 robotic cells** with TCP/IP and Profinet, ensuring reliable multi-sensor communication.
- Designed and 3D printed custom fixtures for robot endurance tests.
- Delivered **5+ onboarding sessions and 3 colloquium talks**, accelerating new-hire integration and knowledge transfer.

August 2022 – March 2023
Fraunhofer IPA, Stuttgart, Germany

Master Thesis – Force Control with Mobile Manipulator

- Developed a digital twin of a mobile manipulator in Gazebo to test autonomous navigation and manipulation.
- Integrated joint- and Cartesian-space modules into a force-controlled pipeline, enabling compliant manipulation.
- Validated across **100+ trials**, achieving **>95%** navigation success and consistent force-controlled manipulation in variable environments.

January 2019 – July 2019
Pennar Industries Ltd, Chennai, India

Quality Control Engineer
- Improved inspection throughput from **12 to 14 checks/day**.
- Reduced railway wagon part rejection rate by 7% through root-cause analysis and process improvement.
- Standardized inspection templates to minimize oversize/undersize deviations.

CERTIFICATIONS & TRAININGS

2025 **AWS Cloud Practitioner Essentials**, Amazon

2024 **Industrial Robotics Programming**, Universal Robots Academy

2022 **ROS2 for Beginners**, Udemy

2021 **Neural Networks and Deep Learning**, Coursera / deeplearning.ai

2021 **Python Bootcamp**, Udemy

ACCOMPLISHMENTS

2025 **Technical Seminar Presentation**
- Led a **team of 5 members** and successfully built an end-to-end pipeline for predictive maintenance of wind turbines and presented it in a University tech seminar.

2023 **Leadership & Team Management**
- Successfully organized a 2-day inter-university cricket tournament, leading a team of volunteers and managing logistics, scheduling, and coordination for **6 participating teams**.

2022 **Cross-Cultural Mentorship**
- Served as a mentor in the Student Buddy Program for **10+ international students**, coordinating with them to facilitate their arrival, orientation, and cultural integration in Germany.

KNOWLEDGE & SKILLS

Core Robotics	ROS | ROS2 | MoveIt | MoveBase | Gazebo | Rviz | ros_control
Programming & AI/ML	Python | C++ | MATLAB | scikit-learn | TensorFlow | OpenCV
Industrial Automation	PLCs (TIA Portal, RSLogix, Simatic S7) | ABB RobotStudio
Simulation & CAD	RoboDK | MuJoCo | CATIA | SolidWorks
Tools & DevOps	Git | Docker | AWS | GitLab CI/CD | Jira | Pytest
Languages	English (C2 - Fluent) | German (B1 - Intermediate)
Soft skills	Adaptability | Cross-functional teamwork | Agile collaboration

RESEARCH INTERESTS

- **Intelligent Robotics:** AI-driven motion planning and Reinforcement learning for autonomous systems.
- **Human–Robot Collaboration:** Shared autonomy and Safe interaction in dynamic environments.
- **Digital Production and Industrial AI:** Digital twins and Data-driven optimization for smart manufacturing.
- **Computer Vision and Perception:** Sensor fusion and Deep learning for robotic applications.

Stuttgart, 4.6.2026

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